Driving the Data Curve: An Analysis and Prediction of the Formula 1 Post-Hybrid Epoch

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1 Abstract

The Formula 1 (F1) Championship, characterized by its blend of cutting-edge technology, elite driving talent, and race dynamics filled with uncertainty, represents the very pinnacle of motorsport. In this paper, we discuss how we can build a novel machine learning model to predict the positions at race finish for every driver. Leveraging data from the post-hybrid era (2022-) of the sport, we explore the predictive capabilities of different algorithms and how the numerical and nominal classifications can contribute to the prediction of points a driver will get.

2 Introduction

Crowned with wreaths, media attention, millions of dollars, and baths of expensive champagne, the winners of Formula 1 races have historically lived the lives many can only hope to witness once in their lifetimes. In a sport that routinely comes down to hundredths, thousandths, and ten-thousandths of a second, the modern rendition of the game has seen a heavy emphasis on data and the importance of leveraging it to its maximum potential. Anything from the speed gained by using the drag reduction system down a strait to the turn angle of a driver's steering wheel is collected and stored for precise analysis.

This paper attempts to utilize machine learning techniques to predict future race winners and fastest drivers. Often correlated but not entirely equivalent, the fastest driver is the driver that completes a single lap of the Grand Prix in the shortest time whereas a race winner is simply the driver that crosses the finish line first. As the only two ways to earn points on the last day of each race weekend, an accurate prediction for the final positions can help each constructor and driver understand if their current strategy or car setups are on track to achieve maximum points and finish on the podium. While we don't have any technical car specifics publicly available, it'll prove to be difficult to simulate strategy effectiveness and forecast factors like tire degradation and fuel consumption. Therefore, this paper aims to serve as a general prediction based off of past race results at the time of data collection. By being able to predict the final race positions for different circuits, we'll be able to predict the winner of the 2024 Driver's Championship (ie. driver with the most points at the end of the season) and Constructor's Championship (ie. team with most points at the end of the season).

The following sections will discuss past related works that cover similar topic domains and explore different analysis techniques. A later discussion on our dataset will provide a more specific scope of our project and the necessary background information to understand the development and test datasets used for this project.

3 Related Work

A similar research work, *A Data-Driven Analysis of Formula 1 Car Races Outcome* by Patil, Jain, et al. [1], dives deeper into factors at play that can affect the outcome of a race. While there is a saying that F1 races are won at the factories before the track, there are always unforeseen factors like the weather that can prove to be the greatest foe of any spectacular driver out on track. In order to identify the factors that are the most important to gaining points, Patil, Jain, et al. decided to perform a correlation analysis as well as a principal component analysis (PCA) to maximize the underlying variance on a lower dimension.

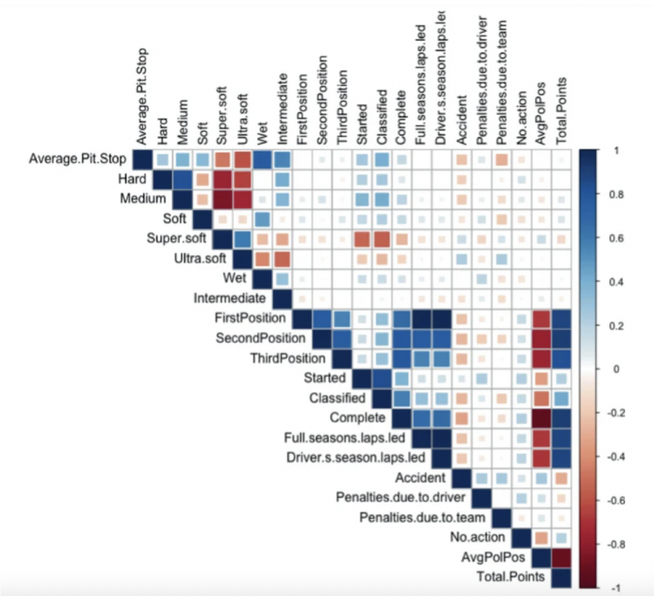

Figure 1: Correlogram of Factors in Patil, Jain, et al (2023)

From Figure 1, we can observe the different factors that the paper had within its dataset, ranging from the tire types (ie. hard, medium, soft, super soft, ultra soft, wet, intermediate) to full seasons laps led to average pole position. The correlogram illustrates that the more teams utilize the super or ultra-soft tires, the more likely a driver is to be penalized. Additionally, the logic behind the correlogram checks out as the higher the average pole or qualifying position, the lower the chance a driver finishes the drive in first, second, or third place. Those positions are also heavily related to the number of laps led in the season, the number of laps completed, and whether the driver finishes more than 90% of the race, defined as "classified" by the authors.

Beyond the interdependencies shown in the previous figure, the authors plotted past race results from the 2015-2019 seasons in a new subspace representation of the first two principal components, along with vectors that correspond to the different race variables. The bi-plot illustrates the aforementioned relationships but also highlights how the first, second, and third positions have a strong contribution to the second principal component or the horizontal axis. Other variables are shown to play a major part in the first principal component, or the vertical axis, potentially explaining the positioning of our drivers. From the plot, we can also observe that penalties and accidents because of the team are highly related.

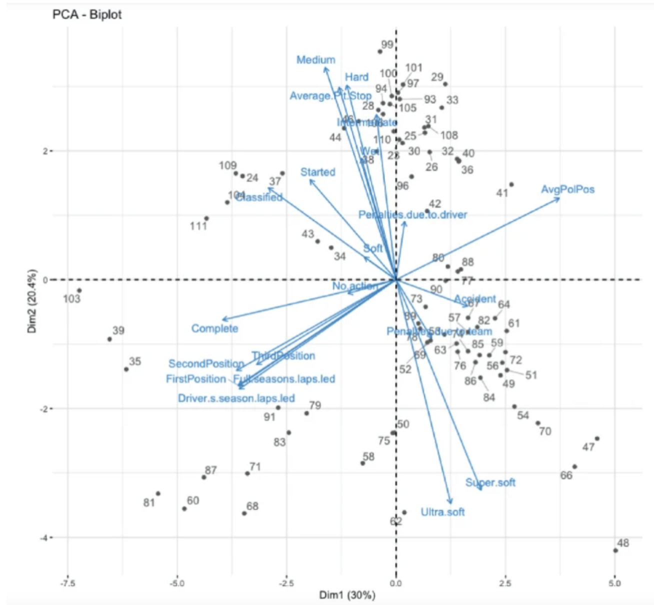

Figure 2: Bi-plot of Principal Components in Patil, Jain, et al (2023)

After these analyses, the authors ran a linear regression model on the data from the 5 selected seasons, where the dependent variable in the model is the total points scored by a driver in each season. The regression model achieved an R-squared value of 99%, meaning the model was able to capture 99% or more of the underlying variance. From the regression, the number of races completed by a driver has a significant impact on the total points scored. In addition, the regression reveals that only medium, soft, ultra-soft, and intermediate tires have a significant impact on the dependent variable. Logically, it made sense the regression showed that the more time and laps a driver spent in the second or third position, there would be positive, notable impact on the final points scored for the season. Lastly, the model was able to translate the inverse relationship previously mentioned about the average qualifying position numerically, with a 3-point decrease in total points scored for every unit increase in the qualifying position.

Another interesting sample addresses the frequently asked question of which to attribute for a race win: the driver, the car, or both? Unlike other "spec" series like IndyCar, Formula 1 stands out as the competition is extended on the driver and construction fronts. *Bayesian Analysis of Formula One Race Results: Disentangling Driver Skill and Constructor Advantage* by Erik-Jan van Kesteren and Tom Bergkamp models a novel Bayesian multilevel rank-ordered logit regression method on the hybrid era of Formula 1 (2014-2021) [2]. The aim of the paper is to scientifically answer the following questions: (a) what is the relative influence of the driver and constructor on race results, (b) how do drivers rank in terms of skill level, and (c) how do constructors rank in terms of race car advantage?

For the model, the outcome variable from the model assumes for each race, there is a set of competitors and outputs ranking of said competitors following a Ranked-Ordered Logit (ROL) distribution based on the latent ability of each competitor. The model is constructed in a manner such that an average driver at an average constructor with an average seasonal performance form will have about a 50% chance of beating a randomly selected other competitor. The model also includes parameters like a seasonal form as well as a long-term form for each driver and constructor. These parameters enable the model to compute seasonal deviation from the long-term performance or advantage a driver and constructor can have. Consequently, the model assumes that a driver's technical skills are independent of the constructor the driver is currently under, meaning a transfer between teams wouldn't affect the driver's skills.

On top of the model, our authors kept in mind that factors like weather can have a serious impact on one's performance in a race as seen in Patil, Jain, et al. However, a fundamental variable not accounted for previously was the design philosophy each constructor has. For instance, Red Bull Racing implores a more high-downforce concept car, more advantageous in narrow yet curvy circuits and less so in fast, strait-heavy circuits. Thus, the authors have four total models: (a) a basic model, (b) a weather model, (c) a circuit type model, and (d) a weather and circuit type model. Through leave-one-out (LOO) cross-validation, the circuit-type model displayed the best performance in terms of expected log posterior density (ELPD) despite a similar performance overall across all four models.

After obtaining posterior samples from Hamiltonian Monte Carlo sampling with 8 chains of 1250 samples and 1000 burn-in iterations, van Kesteren and Bergkamp produced posterior predictive checks (PPCs) for the 2015 and 2019 respective seasons to simulate the early and late stages of the hybrid era. These PPCs act as a good indicator of agreement between the observation and model predictions.

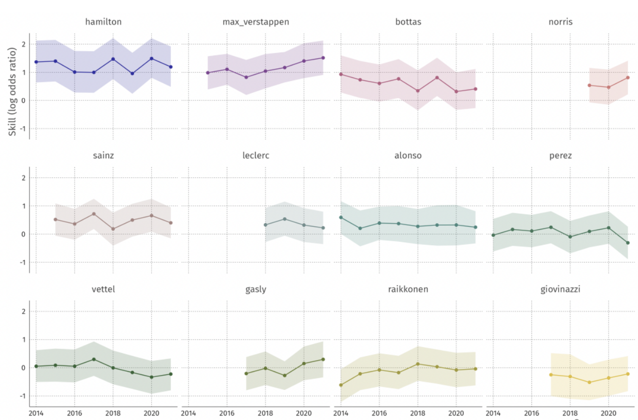

Figure 3: Driver skill trajectories for 12 drivers on the log odds ratio scale in van Kesteren, Bergkamp (2023)

After computing posterior means and credible intervals, the model, accounting for the yearly constructor advantages, indicates that there are slight skill deviations amongst our 12 chosen drivers through the different seasons. Notably, Verstappen, Norris, and Gasly tend to have a skill increase whereas Bottas displays a slight decline. Alonso and Sainz are consistently above the average in latent skills whereas Hamilton remains at the top, consistent with the 6 out of 8 World Driver Championship wins within the hybrid era.

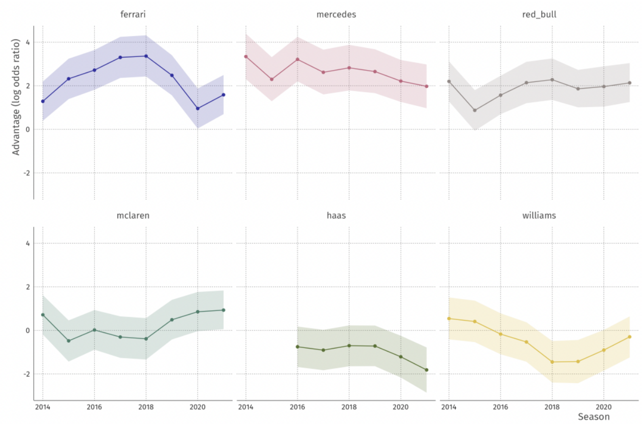

Figure 4: Yearly constructor advantages in the period 2014 – 2020 with 89% credible interval ribbons in van Kesteren, Bergkamp (2023)

Figure 4 clearly illustrates the drop in form from Ferrari between 2019 and 2020 seasons as allegations arose that the Ferrari engine hadn't met league regulations after the 2019 season [3]. This decline consequentially reflected an increase in McLaren's performance in the World Constructor's Championship as we saw the team replace Ferrari for a third-place finish. The paper concluded that the constructors have a larger impact on race results than the driver. In terms of constructors, Mercedes, Ferrari, and Red Bull outperform the rest by a margin while Hamilton and Verstappen clearly stand above the other drivers.

While both cited papers utilize machine learning techniques within the same topic domain and transform related data as this research, both of the papers have yet to explore the domain of predicting future results, the ultimate goal of this paper. It's also noteworthy that the cited papers use data from the hybrid era or earlier, yet the concluding results from each respective paper still apply to the post-hybrid era. The knowledge and results gained from the papers can greatly inform us of the race factors, whether it's the constructor philosophy or the weather, that affect the results we predict.

Track	Position	No.	Driver	Team	Starting Grid	Laps	Time/Retired	Points	+1 Pt	Fastest Lap
Bahrain	1	16	Charles Leclerc	Ferrari	1	57	1:37:33.584	26	Yes	1:34.570
					⋮					
Brazil	2	44	Lewis Hamilton	Mercedes	2	71	1.529	18	No	1:13.942
					⋮					
Spain	18	20	Kevin Magnussen	Haas	17	65	-1	0	No	1:18.069

Table 1: Modified Dataset for the 2022 and 2023 Formula 1 seasons and also first two races of the 2024 season

4 Exploring Our Dataset

Ever since the beginning of the 2022 Formula 1 Season, the illustrious sport's governing body, the FIA, introduced new changes each team constructor must obey. The motivation behind such changes was to promote "closer racing," an attempt to dethrone the historic winning streak set by Lewis Hamilton and Mercedes AMG Petronas. Citing a loss of downforce in the trailing car by dirty air, the new design of cars is created around the principle of using the ground effect to create downforce. As described, the ground effect is an aerodynamic drag reduction that an aircraft's wing, or in this case those of a Formula 1 car, can generate when close to a fixed surface to create a lifting or floating effect [4]. This reintroduction of ground effect since its initial ban in the 80s marks a new epoch in the motorsport. Therefore, for my project, I chose to neglect any past data prior to the 2022 season as a different car setup may introduce bias that doesn't accurately reflect how race results of the near future may turn out. The following is an explanation of our dataset's features and how it was represented in a tabular form:

Track- which circuit the race is held or which grand prix it is
Position- what position the driver finished the race in. Drivers that did not finish the race (DNF) will have a flag of 0
No- the driver's registered number
Driver- the driver's name
Team- which team the driver is competing for in the constructor's championship
Starting Grid- what position the driver will be starting the race from, usually earned through a qualifying session
Laps- laps completed by the driver before DNF or race finish
Time/Retired- the gap between the first car to finish that race. Drivers that won the race will have their final time instead of a gap time. Drivers that did not finish (DNF) will have a flag of -1.
Points- number of points the driver earned from the grand prix
+1 Pt- a binary flag indicating whether or not the driver had the fastest lap of the entire race
Fastest Lap- the driver's fastest lap time completed during the grand prix

Something noteworthy about our dataset is the inclusion of Sprint Race results. We appended the results of the selected Sprint weekends where short stints of about 100 kilometers or a third of the actual race with no mandatory pit stops and tire changes. The Sprint events are scattered across each season calendar, with winners of the Sprint also getting 8 points with a sliding scale down to one point for eighth place. [5]

5 Baseline Performance

Our dataset is an organized CSV file from Github user toUpperCase78 that scrapped the official Formula 1 website for the results of the past few seasons [6]. If the goal is to predict the winner of the 2024 season, we must learn to predict the standings of each race. The following are some baseline tests with 10-fold cross-validation that I ran in Weka on the positions feature, predicting the positions in which each individual driver would end in for every race.

Classifier	Corr. Coeff.	Mean Absolute Error
Additive Regression	0.7867	2.6083
Gaussian Process	0.7966	2.7425
KStar	0.7514	2.4486
RandomForest	0.7103	3.9326
RegByDiscretization	0.8211	1.8992
SMOReg	0.7847	2.9254

Table 2: Positions Baseline Test Results From Original Data

From Table 2, we can observe that Regression by Discretization has the highest correlation coefficient while maintaining the lowest mean absolute error.

However, we must remember that drivers can also earn points by achieving the fastest lap of the race. Thus, we ran another baseline test with 10-fold cross-validation for the +1 Pt. feature.

Classifier	% Correct	Kappa
AdaBoostM1	97.0883	0.4186
DecisionTable	98.1802	0.6973
J48 Tree	98.3621	0.735
JRip	98.7261	0.8045
LMT Tree	98.0892	0.6949

Table 3: +1 Pt. Baseline Test Results From Original Data

From Table 3, we can observe that JRip is the best classifier to accurately forecast our fastest driver per race, as it has the highest correct instances while keeping our kappa statistic high. This indicates that the JRip algorithm enables a substantial agreement between the true and predicted +1 Pt. labels.

In order to improve performance, I changed certain flags and implemented some data cleansing techniques for some identified problematic features. I adjusted the data to have a -1 flag if retired under the Time/Retired feature and also changed the position feature data to have a 0 flag for retired or DNF cars. Similarly, DNF cars will have a 0 flag for the fastest lap as they have no time. We noticed that through our past analysis of our data as well as just observing our data, we could tell that the model is currently a little confused when fed a -1 value for the position feature whenever a driver somehow has not finished the race due to a crash or mechanical failure. Other problematic features are similar, where we also had an N/A value for the Time / Retired feature, representing the time difference with the race winner's time. The regression and other models ran at the moment aren't factoring in the -1 values correctly as the dataset wants, thus technically reducing how useful those data points are. Another problematic feature that visualizing our data highlighted was that we have data across multiple seasons, which team names have changed slightly due to title sponsors (ie. "Red Bull Racing RBPT" vs "Red Bull Racing Honda RBPT"). This is confusing the model more as it treats them as separate entities, which can be a helpful insight but probably not necessary for our tasks at hand. The following are the performances of our best algorithms after the implementation of our changes.

Classifier	Corr. Coeff.	Mean Absolute Error
Additive Regression	0.7942	2.5205
Gaussian Process	0.8004	2.6454
RegByDiscretization	0.831	1.8489

Table 4: Positions Baseline Test Results From Modified Data

Note that as there were no changes in the +1 Pt. feature, our % Correct and Kappa statistics remain the same. From Table 4, we can see that the Regression by Discretization algorithm remains the best algorithm. However, we can see that our performance for each algorithm has increased after our modifications to the original dataset. Thus, for our project, we selected the Regression by Discretization algorithm for the positions feature predictions and the JRip algorithm for the +1 Pt. predictions.

6 Optimization

The next stage is to optimize my selected algorithms to find the best parameter settings to generate the best performance. For our position feature prediction, I decided to optimize is the number of bins for discretization. With a default number of 10 bins, we ran different algorithms on a ten-fold LOO cross-validation on a range of 2 to 25 bins. The optimal parameter was 17 bins, which achieved an average correlation coefficient of 0.8361 and an average mean absolute error of 1.5923. When compared to our baseline tests, we can see that there's a slight improvement of 0.0051 in our correlation coefficient but a slight improvement of 0.2566 in our mean absolute error. For our +1 Pt. feature prediction, I tuned the number of optimization runs for the JRip algorithm, once again using a ten-fold LOO cross-validation methodology. I initially tuned optimization runs from a range of 2, the default, to 10, but my initial tests indicated that 10 performed the best. As I kept running different tuning ranges, I landed on the optimal parameter was 50 runs. Supplying 50 optimization runs in our JRip algorithm, we achieved 99.0901% correct and a kappa statistic of 0.9099. This means that we have notably only 10 misclassification instances out of a total of 1099 instances. Compared to the baseline tests, we had an increase of 0.364% in correct classifications and an increase of 0.1054 in kappa, indicating a better agreement between the predicted and true labels.

7 Final Results

Because of the way we have our development dataset set up, we won't have a lot of data like fastest lap times, starting grid positions, time/retired, and many more data points until the day before any given race. However, since the data collection phase, the 2024 season has seen 6 nail-biting races, meaning we can compare the 4 races not seen in our development data to compare and determine how well our model is. Comparing the test set with our optimized Regression by Discretization model, we get a mean absolute error of about 0.9884. This indicates that on average, our model is about one position away from the true race finish. However, if we were to round our numerical predictions, our model accurately predicted the final race positions of 46 out of 79 instances, with only 13 instances being 1 position away from the true race finish. For our optimized JRip model, we achieved 97.4684 % accuracy as we only had two incorrect predictions for the +1 Pt feature. Contextually, it meant that our model was almost spot-on when it came to predicting which driver was the fastest in any given race. The two incorrect forecasts were seen as surprising results as it's rare for Max Verstappen to not have the fastest lap of the whole race. Therefore, our model was unlikely to predict Fernando Alonso or Oscar Piastri would be able to earn that extra point, seeing they've never gotten it more than twice each in our training set.

By combining the predictions for positions and additional points, our model accurately forecasts the leading three drivers in the Driver's Championship in order: Max Verstappen, Checo Perez, and Charles Leclerc. From the numerical predictions, our discretization regression model is always correct in predicting the second, third, fourth, and eighth-place finishes for each race. While it hadn't predicted the upset win by Lando Norris in the latest Miami Grand Prix, it held the general prediction that Max Verstappen would continue his dominance over the grid. Therefore, Lando is currently above his predicted standings while Carlos Sainz is consequently below his forecast.

8 Discussion

From this project, I think I was able to learn firsthand how much formatting and data processing can affect our performance. Throughout the course, we also learned about identifying and treating problematic features but never at the scale of this training and testing dataset. We can see in section 5 of this paper that our baseline performance increased noticeably once we treated the problematic features by adjusting the flags and cleaned our data by organizing the constructor names. This project also taught me the value of reading related research papers as I never knew a lot of my own questions I had before starting this project existed in papers already publicly available, backed with scientific and mathematical methodologies that aren't entirely above my technical knowledge. It was fulfilling comparing my own methodologies in the processing and analyzing of the discretization regression and JRip algorithms to those of the cited papers. That research illustrated different avenues that my interests of both Formula 1 and machine learning can come together fruitfully. For instance, I had just learned more about principal component analysis in one or two courses, but seeing how the factor vectors can be imposed on the racing data points in a low-dimensional representation certainly connected dots in my own head.

It was honestly rewarding to see the accuracy behind my model as this sport is extremely susceptible to unforeseen variables like the weather, accidents, and mechanical failures. Having a mean absolute error below 1 meant that our model inherently accounted for the possibility of those unforeseen variables without the implementation of a probabilistic model as we see in Van Kesteren, Bergkamp (2023) [2]. My hope for the future is to build on top of this model and be able to predict qualifying positions, which then can be fed to this existing model. Then, I would be able to have a full but naive prediction of the season as a whole. It's a shame that I couldn't have predicted the whole season as I initially wanted for this project due to the setup of our data and model, but I'm hoping that as my machine learning journey progresses, I can begin to include more models for different existing features and new factors like those mentioned in Patil, Jain, et al.

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